Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning.

Albert Gallego Jiménez

Area 4: Pattern Recognition and Computer Vision in
- Geographic Information Systems, satellite images an

MSc in Data Science

Name of the tutor:
Carlos Cabezas Rabadán and Joaquín Torre

Name of the SRP:
David Masip Rodó

March 13, 2026

Universitat Oberta
de Catalunya

Universitat Oberta de Catalunya



This license lets others distribute, remix, adapt, and build upon your work, even commercially, as long as they credit you for the original creation. This is the most accommodating of licenses offered. Recommended for maximum dissemination and use of licensed materials.
https://creativecommons.org/licenses/by-nc/3.0

Final Work Card

Title of the work:	Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning.
Name of the author:	Albert Gallego Jiménez
Name of the tutor:	Carlos Cabezas Rabadán and Joaquín Torres Sospedra
Name of the SRP:	David Masip Rodó
Date of delivery:	March 13, 2026
Studies or Prgram:	MSc in Data Science
Area or the Final Work:	Area 4: Pattern Recognition and Computer Vision in Applied Sciences - Geographic Information Systems, satellite images and geopositioning
Language of the work:	English
Keywords:	Earth Observation, Satellite imagery, Coastal Remote Sensing, Semantic Segmentation, Spectral Indices, Water Segmentation

Abstract

A maximum of 250 words, detailing the purpose, context of application, methodology, results and conclusions of the work.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

This template is conceived as a guide for the student. It can be adapted to the needs of each work, provided that the tutor of the work agrees.

1.1 Context and justification of work

Work starting point (What is the need to cover? Why is it a relevant issue? How is the problem solved at the moment?) and contribution made (What result do you want to obtain?).

It is important to bear in mind that the final work must be understandable for anyone who knows the area of knowledge, but does not have to be an expert in the subject of what the work is about.

1.2 Work objectives

List of objectives of the work.

1.3 Impact on sustainability, ethics-social and diversity

This section should identify the positive and/or negative impacts of the final project on the three dimensions of UOC's cross-disciplinary competence "Ethical and global commitment".

The Cross-cutting Guide on Ethical and Global Competence will help you write these sections.

1.4 Focus and method followed

Mention of what are the possible strategies to carry out the work and what is the chosen strategy (develop a new product, adapt an existing product...). It is necessary to include an assessment of why this is the most appropriate strategy to achieve the objectives.

1.5 Work schedule

Description of the necessary resources to do the work, the tasks to be carried out and a temporary planning of each task using a Gantt diagram or similar. This planning should mark the partial milestones of each of the CAPs.

1.6 Brief summary of products obtained

No need to go into detail: the detailed description will be made in the rest of the chapters.

1.7 Brief description of other memory chapters

Brief explanation of the contents of each chapter and its relationship with the global project.

Chapter 2

Materials and methods

In these sections, it is necessary to describe:

- The most relevant aspects of the design and development of the work.
- The methodology chosen to carry out this development, describing the possible alternatives, the decisions taken, and the criteria used to make these decisions.
- The products obtained.

Chapter 3

Statement on the use of generative artificial intelligence

Explanation of any use made of generative artificial intelligence tools (such as ChatGPT, Gemini, DALL-E, or others) during the conception and development of this Final Project, as well as the writing of this report and the preparation of the presentation. It is mandatory to specify the concrete tools used, the purpose (e.g., ideation, code outlining, grammatical correction, image generation, etc.), the periods or phases of the work in which they were employed, the actions taken to validate the content generated by AI, and a critical evaluation of the impact of their use. It must be explicitly clarified which parts of the work are entirely the student's authorship and which have been based on inputs or results generated by AI, ensuring at all times that the intellectual responsibility and final validation of the content fall entirely upon the student, and that its use has been discussed with the Project supervisor and carried out in accordance with their instructions.

Chapter structuring may vary depending on the type of work.

If applicable, a section on "Economic evaluation of work" will be included. This section will indicate the expenses associated with the development and maintenance of work, as well as the economic benefits obtained and a final analysis on the viability of the product.

Chapter 4

Results

Detail in this section the results obtained using the methodology described in the previous section.

The figures must be explained and quoted in the text, such as **??**, in which the error is shown depending on the distance, in arbitrary units. All graphs must have the title of the axes.

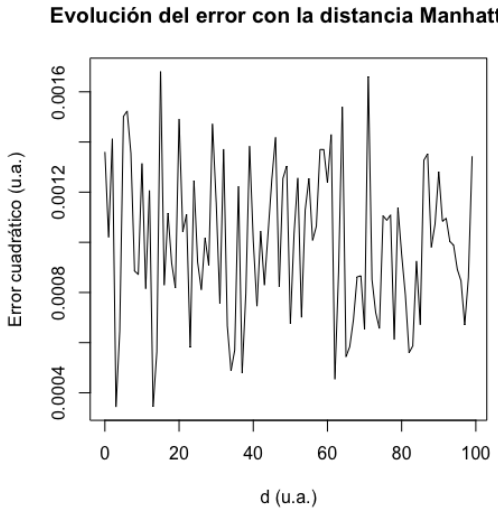


Figure 4.1: Error working distance in arbitrary units.

Chapter 5

Conclusions and future works

This chapter must include:

- A description of the conclusions of the work:
  - Once the results have been obtained, what conclusions are drawn?
  - Are these results as expected? Or have they been amazing? Why?
- A critical reflection on the achievement of the objectives initially set:
  - Have we achieved all the objectives? If the answer is negative, why?
- A critical analysis of the monitoring of planning and methodology throughout the product:
  - Has the schedule been followed?
  - Has the planned methodology been adequate enough?
  - Has it been necessary to introduce changes to ensure the success of the work? Why?
- From the impacts foreseen in **??**, ethical-social, sustainability and diversity, evaluate/mention whether they have been mitigated (if they were negative) or whether they have been achieved (if they were positive).
- If unforeseen impacts have appeared in **??**, evaluate/mention how they have mitigated (if they were negative) or what they have contributed (if they were positive).
- The future lines of work that have not been explored in this work and have remained pending.

Chapter 6

Glossary

Definition of the most relevant terms and acronyms used in the Report.

Chapter 7

Bibliography

Numbered list of bibliographic references used within memory. In each place where a reference is used within the text, it must be indicated by quoting the reference number, for example: [7].

It is very important to include all the references used and quote them appropriately, that is, including all the information necessary to identify the reference. The minimum information to be included according to the reference type is:

- Book: Authors, Title, Edition (if applicable) Editorial, City, Year.
- Journal article: Authors, Title, Name of the Journal, Home and End Page Number, Issue of the magazine / Volume, Year.
- Web: URL and date it was visited.

Information on how to quote documents: `http://biblioteca.uoc.edu/es/recursos/citacion-bibliografica`.

List of sections that are too extensive to include in memory and have a self-contained character (e.g. user manuals, installation manuals, etc.)

Depending on the type of work, there may be no need to add an annex.